

CHARLES GUILLE-ESCURET

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EDUCATION

Mila, Quebec AI Institute and Université de Montréal (UdeM) <i>Doctor of Philosophy in Computer Science; GPA 4.0/4.0</i>	<i>Montréal, Canada</i> 2019 — 2025
École Normale Supérieure Paris-Saclay <i>Master of Science in Applied Mathematics - MVA program ; cum laude</i> <i>Bachelor of Computer Science; cum laude</i>	<i>Paris, France</i> 2016 2014

EXPERIENCE

Independent Research Consultant – LLM pretraining • Advised pretraining of Mixture-of-Experts with hundreds of billions of parameters with Megatron. • Advised adoption and implementation of next generation optimizers such as Muon for trillion-parameters training runs.	<i>Montréal, Canada</i> July 2025 - Now
ServiceNow, Foundation Model Lab <i>Visiting Researcher</i> • Led the pretraining of a 7b-parameters language model over 1.3T tokens. • Investigated domain-specific data retrieval and validation pipelines for internal use cases.	<i>Montréal, Canada</i> May - Dec. 2024
Apple MLR <i>Research Intern</i> • Designed a novel method to construct finite-sample valid confidence sets for linear regression in a much more realistic setting than previous works, with a wide range of applications. • Incorporated conformal predictions to previous method to further improve performances and computation costs.	<i>Paris, France</i> Apr. 2023 - Sept. 2023
ServiceNow, ATG Research Group <i>Visiting Researcher, Low-data learning team (part-time)</i> • Trained self-supervised contrastive models to perform anomaly detection on challenging benchmarks with a novel method. • Proposed a novel benchmark for broad OOD detection and showed how to alleviate the inconsistencies of recent works by ensembling.	<i>Montréal, Canada</i> Dec. 2021 - May 2022 and Sept 2022 - Feb 2023
Mila Quebec AI Institute and Université de Montréal (UdeM) <i>Doctoral Candidate, advised by Prof. Ioannis Mitliagkas</i> • Research Areas: <i>First-order optimization for deep learning, learning dynamics, LLM pretraining.</i> • Teaching assistant for 3 classes and co-supervised 3 internships. • Introduced a new theoretical framework for the study of first-order optimization. • Derived novel upper and lower bounds on the convergence rate of first-order algorithms. • Contributed to various projects on applications such as insect classification, lightning current prediction, predictive visualization of flood effects, adversarial robustness.	<i>Montréal, Canada</i> 2019 – 2025
University of California Berkeley <i>Visiting Scholar, EECS department, under the supervision of Prof. Alexandre Bayen</i> • Worked with startup <i>SafelyYou</i> to implement then-SOTA computer vision models for automatic fall detection in elderly care facilities. • Resulted in a 80% decrease in ER visits and 40% decrease in fall frequency.	<i>Berkeley, CA</i> Oct. 2017 – Oct. 2018
Elum Energy <i>Machine Learning Engineer</i> • Built from scratch all predictive models for energy consumption. • These models were successfully deployed to optimize battery usage. • Conducted interviews, designed and evaluated tests for ML recruitment.	<i>Paris, France</i> Sept. 2016 – Mar. 2017
National Institute of Informatics <i>ML Research Intern</i> • Designed and implemented a method to infer SPARQL queries from question-answer pairs.	<i>Tokyo, Japan</i> Apr. 2016 – Aug. 2016

HONORS AND AWARDS

Top reviewer AISTATS top 10% of reviewers	2023
Best Student Paper Award NeurIPS OPT2020, for "A Study of Condition Numbers for First-Order Optimization"	2020
Data Intelligence Award - Predictive Analytics Data Intelligence Forum Paris, for work at Elum Energy	2016
Scholar Award , NeurIPS 2022	2022
Normalien Fellowship 4 years merit-based fellowship	2013

TEACHING EXPERIENCE

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- Mila Optimization Crash Course, Lecturer** Spring 2024
 • Gave lectures on Adam and RMSProp.
- Mila and UdeM, IFT3395 and IFT6390 Fundamentals of Machine Learning, Teaching Assistant** Fall 2019
 • Designed and taught Python workshops for Machine Learning.
 • Contributed to writing assignments and exams and their grading.
- HEC Montreal, MATH80629A ML for Large Scale Data Analysis and Decision Making, Teaching Assistant** Winter 2021
 • Designed and taught Python workshops for Machine Learning, assisted students.
- UdeM, EDUlib, SD1FR MOOC Data Science, Instructor** Jan. - Aug. 2021
 • Developed Jupyter Notebooks for an online MOOC on data science hosted by EDUlib.
- Mila, internship co-supervisor** 2020
 • Academic co-supervisor for 3 students of Mila's professional masters in AI doing their internship in external companies.

OTHER ACADEMIC CONTRIBUTIONS

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- **Co-organizer**, 4th Neural Scaling Laws workshop 2022
 • **Co-organizer**, Montreal Machine Learning and Optimization (MTL ML-OPT) seminars. 2023-now

PUBLICATIONS *: co-first authors

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- Ostapenko O.*, **Guille-Escuret C.***, Kumar L., Tian M., Kocetkov D., Subbaraj G., Li R., Lamy-Poirier J., Paquet S., Scholak T. (2024). Using Scaling Laws for Data Source Utility Estimation in Domain-Specific Pre-Training, *CoLM 2025 Workshop on Multilingual Data Quality Signals*.
- H Zhang T.*, Miligan A.*, Maes L.*, Jolicoeur-Martineau A., Mitliagkas I., Scieur D., Lacoste-Julien S., **Guille-Escuret C.** (2024). Understanding Adam Requires Better Rotation Dependent Assumptions, *NeurIPS 2025*.
- Guille-Escuret C.**, Ndiaye E. (2024). From Conformal Predictions to Confidence Regions, *ArXiv*.
- Guille-Escuret C.**, Ndiaye E. (2023). Finite Sample Confidence Regions for General Linear Regression Parameters Using Arbitrary Predictors, *ArXiv*.
- Guille-Escuret C.***, Naganuma H.*, Fatras K., Mitliagkas I. (2023). No Wrong Turns: The Simple Geometry Of Neural Networks Optimization Paths, *ICML 2024*.
- Guille-Escuret C.**, Noel P., Vazquez D., Mitliagkas I., Monteiro J. (2023). Expecting The Unexpected: Towards Broad Out-Of-Distribution Detection, *NeurIPS 2024*.
- Guille-Escuret C.**, Rodriguez P., Vazquez D., Mitliagkas I., Monteiro J. (2022). CADet: Fully Self-Supervised Anomaly Detection With Contrastive Learning, *NeurIPS 2023*.
- Ibrahim A., **Guille-Escuret C.**, Mitliagkas I., Rish I., Krueger D., Bashivan P. (2022). Towards Out-of-Distribution Adversarial Robustness, *ICML 2022 New Frontiers In Adversarial Machine Learning Workshop*.
- Guille-Escuret C.**, Rodriguez P., Vazquez D., Mitliagkas I., Monteiro J. (2022). Contrastive Self-supervision Defines General-Purpose Similarity Functions, *NeurIPS 2022, Self-supervised learning - theory and practice workshop*.
- Guille-Escuret C.**, Ibrahim A., Goujaud B., Mitliagkas I. (2022). Gradient Descent Is Optimal Under Lower Restricted Secant Inequality And Upper Error Bound, *NeurIPS 2022*.
- Guille-Escuret C.***, Goujaud B.*, Girotti M., Mitliagkas I. (2021). A Study of Condition Numbers for First-Order Optimization, *AISTATS 2021* and best student paper award at *NeurIPS 2020, Optimization for Machine Learning Workshop*.
- Monferran P.*, **Guille-Escuret C.***, Guiffaut C., Reineix A. (2021). Prediction of Lightning Currents on Fastening Assemblies of an Aircraft Fuel Tank with Machine Learning Methods, *IEEE Transactions on electromagnetic compatibility*.
- Boussioux L., Giro-Larraz T., **Guille-Escuret C.**, Cherti M., Kégl B. (2019). InsectUp: Crowdsourcing Insect Observations to Assess Demographic Shifts and Improve Classification, *ICML 2019, Workshop on AI for Social Good* and *ICCV 2019, Workshop on Wildlife Conservation*